CHAPTER 7

© iStock/ShutterStock, Inc.

Flexibility Training Concepts

OBJECTIVES

After studying this chapter, you will be able to:

- ✔ Explain the effects of muscle imbalances on the human movement system (kinetic chain).
- ✔ Provide a scientific rationale for the use of an integrated flexibility training program.
- ✔ Differentiate between the various types of flexibility techniques.
- ✔ Perform and instruct appropriate flexibility techniques for given situations.

Introduction to Flexibility Training

Once all of the assessment data has been collected as described in the Fitness Assessment chapter, the remainder of the programming template can be completed and the focus can be turned to designing the exercise program. The next portion of the Optimum Performance Training™ (OPT™) programming template that needs to be completed is the warm-up section. In designing the warm-up program, the components of flexibility and cardiorespiratory training need should be reviewed. Most clients will require some flexibility training to safely and effectively perform exercise optimally, and that will be the focus of this chapter.

Current Concepts in Flexibility Training

In today's society, nearly everyone is plagued by postural imbalances largely as the result of sedentary lifestyles, advancements in technology, and repetitive movements. Office jobs that require individuals to sit for long hours have led to dramatic increases in work-related injuries, including low-back and neck pain and carpal tunnel syndrome, as well as increased rates of obesity. Flexibility training has become



STRETCH Your Knowledge

Is Lack of Flexibility Associated with Risk of Injury?

Several studies have found an association between altered range of motion (ROM), muscle tightness, or lack of flexibility with increased risk of injury. However, literature reviews have also found studies in which no association was found. More controlled high-quality studies are needed before firm conclusions can be drawn.

- Witvrouw et al. (2003) in a prospective study with 146 male soccer players found that soccer players with increased muscle tightness in the quadriceps and hamstring complex were found to have a statistically greater risk for injury (1).
- Witvrouw et al. (2001) in a prospective study found that decreased flexibility of the hamstring complex and quadriceps significantly contributes to the development of patellar tendonitis in the athletic population (2).
- Cibulka et al. (1998) in a cross-sectional study of 100 patients with unspecified low-back pain demonstrated unilateral hip rotation asymmetry (3).
- Knapik et al. (1991) reported that strength and flexibility imbalances in female collegiate athletes were associated with lower extremity injuries (4).
- Maffey and Emery (2007) found no consistent evidence that adductor tightness increased the risk of groin strains, but also suggested that more research is required before conclusions can be drawn (5).

REFERENCES

1. Witvrouw E, Danneels L, Asselman P, D'Have T, Cambier D. Muscle flexibility as a risk factor for developing muscle injuries in male professional soccer players. A prospective study. *Am J Sports Med.* 2003;31(1):41-46.
2. Witvrouw E, Bellemans J, Lysens R, Danneels L, Cambier D. Intrinsic risk factors for the development of patellar tendinitis in an athletic population. A two-year prospective study. *Am J Sports Med.* 2001;29(2):190-195.
3. Cibulka MT, Sinacore DR, Cromer GS, Delitto A. Unilateral hip rotation range of motion asymmetry in patients with sacroiliac joint regional pain. *Spine.* 1998;23(9):1009-1015.
4. Knapik JJ, Bauman CL, Jones BH, Harris JM, Vaughan L. Preseason strength and flexibility imbalances associated with athletic injuries in female collegiate athletes. *Am J Sports Med.* 1991;19(1):76-81.
5. Maffey L, Emery C. What are the risk factors for groin strain injury in sport? A systematic review of the literature. *Sports Med.* 2007:37(10):881-894.

increasingly recognized as an important way to help aid in preventing and treating various neuromuscular injuries. Clients without adequate levels of flexibility and joint motion may be at increased risk of injury, and may not be able to achieve their personal fitness goals until these deficits are corrected (1–4). It is important for personal trainers to understand the principles of flexibility training to be able to properly design an integrated training program (1–3).

Flexibility The normal extensibility of all soft tissues that allows the full range of motion of a joint.

Extensibility Capability to be elongated or stretched.

What Is Flexibility?

Flexibility can be simply described as the ability to move a joint through its complete range of motion. Range of motion (ROM) of a joint is dictated by the normal **extensibility** of all soft tissues surrounding it (1). An important characteristic of soft

tissue is that it will only achieve efficient extensibility if optimal control of movement is maintained throughout the entire ROM (5). Optimal control of movement throughout a joint's entire ROM is referred to as **dynamic range of motion**. Dynamic ROM is the combination of flexibility and the nervous system's ability to control this range of motion efficiently. There are various factors that can influence flexibility, including:

Dynamic range of motion The combination of flexibility and the nervous system's ability to control this range of motion efficiently.

- Genetics
- Connective tissue elasticity
- Composition of tendons or skin surrounding the joint
- Joint structure
- Strength of opposing muscle groups
- Body composition
- Sex
- Age
- Activity level
- Previous injuries or existing medical issues
- Repetitive movements (pattern overload)

Neuromuscular efficiency is the ability of the nervous system to recruit the correct muscles (agonists, antagonists, synergists, and stabilizers) to produce force (concentrically), reduce force (eccentrically), and dynamically stabilize (isometrically) the body's structure in all three planes of motion.

Neuromuscular efficiency The ability of the neuromuscular system to allow agonists, antagonists, and stabilizers to work synergistically to produce, reduce, and dynamically stabilize the entire kinetic chain in all three planes of motion.

For example, when performing a cable pulldown exercise, the latissimus dorsi (agonist) must be able to concentrically accelerate shoulder extension, adduction, and internal rotation while the middle and lower trapezius and rhomboids (synergists) perform downward rotation of the scapulae. At the same time, the rotator cuff musculature (stabilizers) must dynamically stabilize the glenohumeral (shoulder) joint throughout the motion. If these muscles (force-couples) do not work in tandem efficiently, compensations may ensue, leading to muscle imbalances, altered joint motion, and possible injury.

To allow for optimal neuromuscular efficiency, individuals must have proper flexibility in all three planes of motion. This allows for the freedom of movement needed to perform everyday activities effectively, such as bending over to tie shoes or reaching in the top cupboard for dishes **Table 7.1**. To summarize, flexibility requires extensibility, which requires dynamic range of motion, which requires neuromuscular efficiency. This entire chain is achieved by taking an integrated (comprehensive) approach toward flexibility training.

Flexibility training must use a multifaceted approach, which integrates various flexibility techniques to achieve optimal soft tissue extensibility in all planes of motion Table 7.1. To better understand integrated flexibility, a few important concepts must first be reviewed, including the human movement system, muscle imbalances, and neuromuscular efficiency.

Review of the Human Movement System

The human movement system (HMS), also known as the kinetic chain, comprises the muscular, skeletal, and nervous systems. Optimal alignment and function of each component of the HMS is the cornerstone of a sound training program. If one or more segments of the HMS are misaligned and not functioning properly, predictable patterns

TABLE 7.1 **Multiplanar Flexibility**

© Rotab/ShutterStock, Inc.

Muscle	Plane of Motion	Movement
Latissimus dorsi	Sagittal	Must have proper extensibility to allow for proper shoulder flexion
	Frontal	Must have proper extensibility to allow for proper shoulder abduction
	Transverse	Must have proper extensibility to allow for proper external humerus rotation
Biceps femoris	Sagittal	Must have proper extensibility to allow for proper hip flexion, knee extension
	Frontal	Must have proper extensibility to allow for proper hip adduction
	Transverse	Must have proper extensibility to allow for proper hip and knee internal rotation
Gastrocnemius	Sagittal	Must have proper extensibility to allow for proper dorsiflexion of ankle
	Frontal	Must have proper extensibility to allow for proper inversion of calcaneus
	Transverse	Must have proper extensibility to allow for proper internal rotation of femur

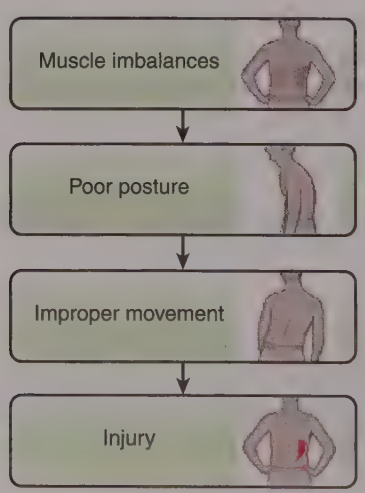
Postural distortion patters Predictable patterns of muscle imbalances.

of dysfunction develop (5–8). These patterns of dysfunction are referred to as **postural distortion patterns**, which can lead to decreased neuromuscular efficiency and tissue overload **Figure 7.1** (5).

Postural distortion patterns (poor static or dynamic posture) are represented by a lack of structural integrity, resulting from decreased functioning of one (or more) components of the HMS (5–7). A lack of structural integrity can result in altered length-tension relationships (altered muscle lengths), altered force-couple relationships (altered muscle activation), and altered arthrokinematics (altered joint motion). There are several postural distortions that personal trainers must be aware of, all of which are reviewed in the Fitness Assessment chapter (chapter six). Maximal neuromuscular efficiency of the HMS can only exist if all components (muscular, skeletal, and neural) function optimally and interdependently. The ultimate goal of the HMS is to maintain homeostasis (or dynamic postural equilibrium).

FIGURE 7.1

Postural distortion patterns.



Poor flexibility can lead to the development of **relative flexibility** (or altered movement patterns), which is the process in which the HMS seeks the path of least resistance, during functional movement patterns (9). A prime example of relative flexibility is seen in people who squat with their feet externally rotated **Figure 7.2**. Because most people have tight calf muscles they lack the proper amount of dorsiflexion at the ankle to perform a squat with proper mechanics. By widening the stance and externally rotating the feet it is possible to decrease the amount of dorsiflexion required at the ankle to perform a squat using good technique. A second example can be seen when people perform an overhead shoulder press with excessive lumbar extension (arched lower back) **Figure 7.3**. Individuals who possess a tight latissimus dorsi will have decreased sagittal-plane shoulder flexion (inability to lift arms directly overhead), and as a result, they compensate for this lack of range of motion at the shoulder in the lumbar spine to allow them to press the load completely above their head.

Relative flexibility The tendency of the body to seek the path of least resistance during functional movement patterns.

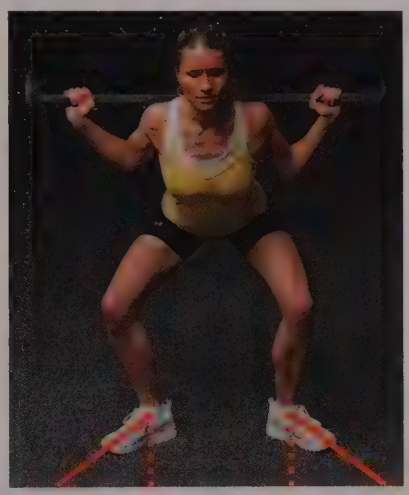


FIGURE 7.2

Squat with externally rotated feet.

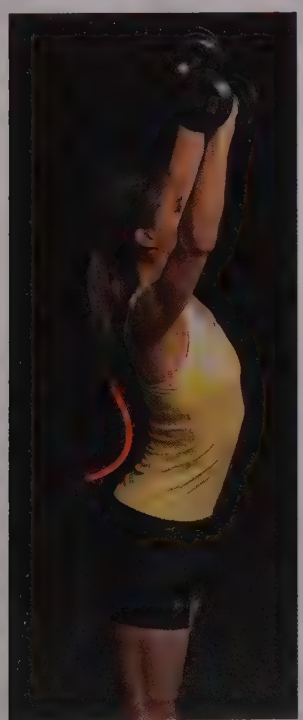


FIGURE 7.3

Overhead shoulder press with lumbar extension.

Muscle Imbalance

Muscle imbalance
Alteration of muscle length surrounding a joint.

Muscle imbalances are alterations in the lengths of muscles surrounding a given joint **Figure 7.4** and **Figure 7.5**, in which some are overactive (forcing compensation to occur) and others may be underactive (allowing for the compensation to occur) (5,7). Examples of such imbalances in the form of movement compensations are discussed in chapter six.

Muscle imbalance can be caused by a variety of mechanisms (1,9). These causes may include:

- Postural stress
- Emotional duress
- Repetitive movement
- Cumulative trauma
- Poor training technique
- Lack of core strength
- Lack of neuromuscular efficiency

FIGURE 7.4

Muscle balance.

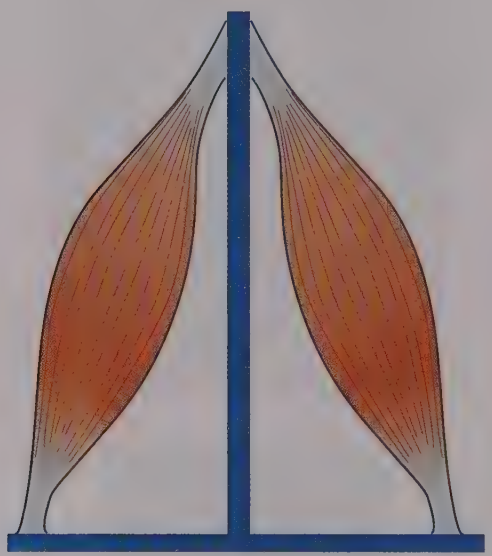
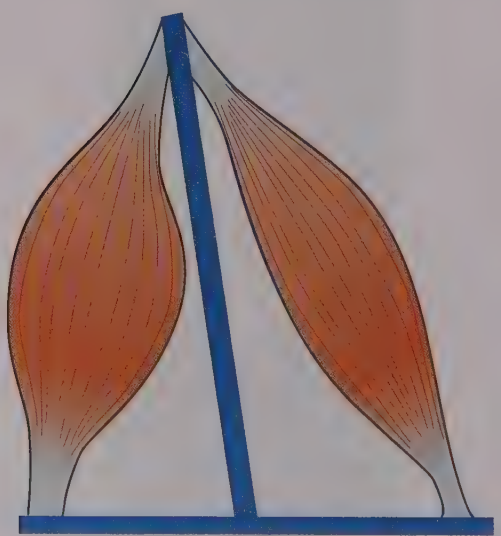


FIGURE 7.5

Muscle imbalance.



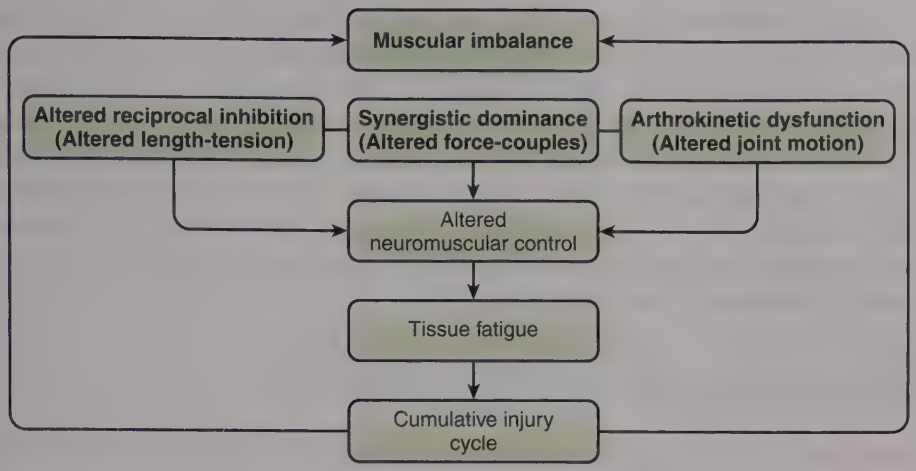


FIGURE 7.6
Muscular imbalance.

Muscle imbalances may be caused by or result in altered reciprocal inhibition, synergistic dominance, arthrokinetic dysfunction, and overall decreased neuromuscular control **Figure 7.6**. These concepts are reviewed below.

Altered Reciprocal Inhibition

Reciprocal inhibition is a naturally occurring phenomenon that allows movement to take place. **Reciprocal inhibition** is defined as the simultaneous contraction of one muscle and the relaxation of its antagonist. For example, to perform elbow flexion during a biceps curl, the biceps brachii actively contracts while the triceps brachii (the antagonist muscle) relaxes to allow the movement to occur. However, **altered reciprocal inhibition** is caused by a tight agonist muscle decreasing the neural drive to its functional antagonist (1,5–7,10–16). For example, a tight psoas (hip flexor) would decrease neural drive of the gluteus maximus (hip extensor). Altered reciprocal inhibition alters force-couple relationships, produces synergistic dominance, and leads to the development of faulty movement patterns, poor neuromuscular control, and arthrokinetic (joint) dysfunction.

Synergistic Dominance

Synergistic dominance is a neuromuscular phenomenon that occurs when synergists take over function for a weak or inhibited prime mover (7). For example, if the psoas is tight, it leads to altered reciprocal inhibition of the gluteus maximus, which in turn results in increased force output of the synergists for hip extension (hamstring complex, adductor magnus) to compensate for the weakened gluteus maximus. The result of synergistic dominance is faulty movement patterns, leading to arthrokinetic dysfunction and eventual injury (such as hamstring strains).

Arthrokinetic Dysfunction

The term **arthrokinematics** refers to the motion of the joints. **Arthrokinetic dysfunction** is a biomechanical and neuromuscular dysfunction leading to altered joint motion (5–8). Altered joint motion can be caused by altered length-tension relationships and force-couple relationships, which affect the joint and cause poor movement efficiency.

Reciprocal inhibition The simultaneous contraction of one muscle and the relaxation of its antagonist to allow movement to take place.

Altered reciprocal inhibition The concept of muscle inhibition, caused by a tight agonist, which inhibits its functional antagonist.

Synergistic dominance The neuromuscular phenomenon that occurs when inappropriate muscles take over the function of a weak or inhibited prime mover.

Arthrokinematics The motions of joints in the body.

Arthrokinetic dysfunction Altered forces at the joint that result in abnormal muscular activity and impaired neuromuscular communication at the joint.

For example, performing a squat with excessively externally rotated feet (feet turned outward) forces the tibia (shin bone) and femur (thigh bone) to also rotate externally. This posture alters the length-tension relationships of the muscles at the knees and hips, putting the gluteus maximus in a shortened position and decreasing its ability to generate force. Furthermore, the biceps femoris (hamstring muscle) and piriformis (outer hip muscle) become synergistically dominant, altering the force-couple relationships and ideal joint motion, increasing the stress on the knees and low back (17). With time, the stress associated with arthrokinetic dysfunction can lead to pain, which can further alter muscle recruitment and joint mechanics (5–7).

Neuromuscular Efficiency

As mentioned earlier, neuromuscular efficiency is the ability of the neuromuscular system to properly recruit muscles to produce force (concentrically), reduce force (eccentrically), and dynamically stabilize (isometrically) the entire kinetic chain in all three planes of motion. Because the nervous system is the controlling factor behind this principle, it is important to mention that *mechanoreceptors* (or sensory receptors) located in the muscles and tendons help to determine muscle balance or imbalance. Mechanoreceptors include the muscle spindles and Golgi tendon organs.

Muscle Spindles

As mentioned in chapter two, muscle spindles are the major sensory organ of the muscle and are composed of microscopic fibers that lie parallel to the muscle fiber. Remember that muscle spindles are sensitive to change in muscle length and rate of length change (5,18–25). The function of the muscle spindle is to help prevent muscles from stretching too far or too fast. However, when a muscle on one side of a joint is lengthened (because of a shortened muscle on the opposite side), the spindles of the lengthened muscle are stretched. This information is transmitted to the brain and spinal cord, exciting the muscle spindle and causing the muscle fibers of the lengthened muscle to contract. This often results in micro muscle spasms or a feeling of tightness (1,5,6).

The hamstring complex is a prime example of this response when the pelvis is rotated anteriorly **Figure 7.7**, meaning the anterior superior iliac spines (front of the pelvis) move downward (inferiorly) and the ischium (bottom posterior portion of pelvis, where the hamstrings originate) moves upward (superiorly). If the attachment of the hamstring complex is moved superiorly, it increases the distance between the two attachment sites and lengthens the hamstring complex. In this case, the hamstring

FIGURE 7.7

Effect of the hamstring complex with an anteriorly rotated pelvis.



complex does not need to be statically stretched because it is already in a stretched position. When a lengthened muscle is stretched, it increases the excitement of the muscle spindles and further creates a contraction (spasm) response. With this scenario, the shortened hip flexors are helping to create the anterior pelvic rotation that is causing the lengthening of the hamstring complex. Instead, the hip flexors need to be stretched (17). (This will be reviewed later in the chapter.)

Another example includes an individual whose knees adduct and internally rotate (knock-knees) during a squat exercise. The underactive muscle is the gluteus medius (hip abductor and external rotator), and the overactive muscles include the adductors (inner thighs) and tensor fascia latae (a hip flexor and hip internal rotator). Thus, one would not need to stretch the gluteus medius, but instead stretch the adductor complex and tensor fascia latae, which in this case are overactive, pulling the femur into excessive adduction and internal rotation.

Golgi Tendon Organs

As also mentioned in chapter two, Golgi tendon organs are located within the *musculotendinous junction* (or the point where the muscle and the tendon meet) and are sensitive to changes in muscular tension and the rate of tension change (5,18–25). When excited, the Golgi tendon organ causes the muscle to relax, which prevents the muscle from being placed under excessive stress, which could result in injury. Prolonged Golgi tendon organ stimulation provides an inhibitory action to muscle spindles (located within the same muscle). This neuromuscular phenomenon is called **autogenic inhibition** and occurs when the neural impulses sensing tension are greater than the impulses causing muscle contraction (14). The phenomenon is termed *autogenic* because the contracting muscle is being inhibited by its own receptors.

Autogenic inhibition The process by which neural impulses that sense tension are greater than the impulses that cause muscles to contract, providing an inhibitory effect to the muscle spindles.



Memory Jogger

Autogenic inhibition is one of the main principles used in flexibility training, particularly with static stretching in which one holds a stretch for a prolonged period. Holding a stretch creates tension in the muscle. This tension stimulates the Golgi tendon organ, which overrides muscle spindle activity in the muscle being stretched, causing relaxation in the overactive muscle and allowing for optimal lengthening of the tissue. In general, stretches should be held long enough for the Golgi tendon organ to override the signal from the muscle spindle (approximately 30 seconds).

SUMMARY

Flexibility training can reduce the risk of muscle imbalances, joint dysfunctions, and overuse injuries. It is important to have proper range of motion in all three planes (sagittal, frontal, transverse), which can be achieved by implementing an integrated approach toward flexibility training. All segments of the kinetic chain must be properly aligned to avoid postural distortion patterns (poor posture) and tissue overload. The adaptive

potential of the HMS is decreased by limited flexibility, which forces the body to move in an altered fashion, leading to relative flexibility (faulty movement patterns).

Muscle imbalances result from altered length-tension relationships, force-couple relationships, and arthrokinematics. These imbalances can be caused by poor posture, poor training technique, or previous injury. These muscle imbalances result in altered reciprocal inhibition, synergistic dominance, and arthrokinetic dysfunction, which in turn lead to decreased neuromuscular control and possibly injury.

Scientific Rationale for Flexibility Training

Flexibility training is a key component for all training programs (1,5). It is used for a variety of reasons, including:

- Correcting muscle imbalances
- Increasing joint range of motion
- Decreasing the excessive tension of muscles
- Relieving joint stress
- Improving the extensibility of the musculotendinous junction
- Maintaining the normal functional length of all muscles
- Improving neuromuscular efficiency
- Improving function

Pattern Overload

Pattern overload
Consistently repeating the same pattern of motion, which may place abnormal stresses on the body.

Muscular imbalances are highly prevalent in today's society and are oftentimes caused by **pattern overload**. Pattern overload is consistently repeating the same pattern of motion, such as baseball pitching, long-distance running, and cycling, which with time places abnormal stresses on the body. There are gym members who train with the same routine repetitively. This too may lead to pattern overload and place abnormal stresses on the body.

Pattern overload may not necessarily be directly related to exercise. For example, a loading-dock employee who has a particularly repetitive occupation lifting and loading packages all day is prone to pattern overload. Even sitting for long periods of time while working on a computer is a repetitive stress.

Cumulative Injury Cycle

Poor posture and repetitive movements create dysfunction within the connective tissue of the body (1,5,26–28). This dysfunction is treated by the body as an injury, and as a result, the body will initiate a repair process termed the cumulative injury cycle **Figure 7.8** (5,28).

Any trauma to the tissue of the body creates inflammation. Inflammation, in turn, activates the body's pain receptors and initiates a protective mechanism, increasing muscle tension or causing muscle spasm. Heightened activity of muscle spindles in particular areas of the muscle create a microspasm, and as a result of the spasm, adhesions (or knots) begin to form in the soft tissue **Figure 7.9**. These adhesions form

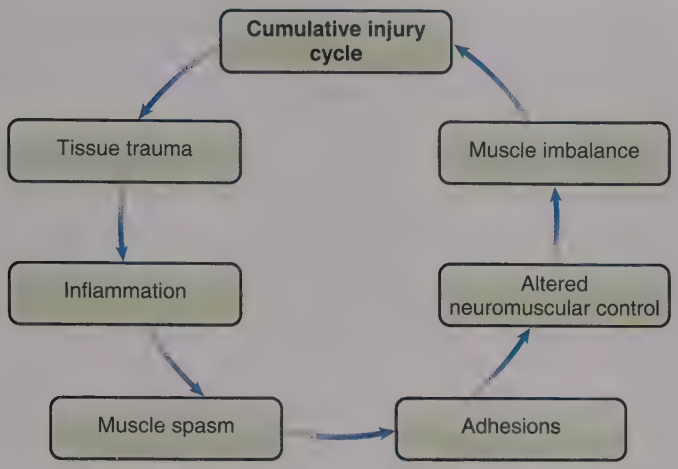


FIGURE 7.8
Cumulative injury cycle.

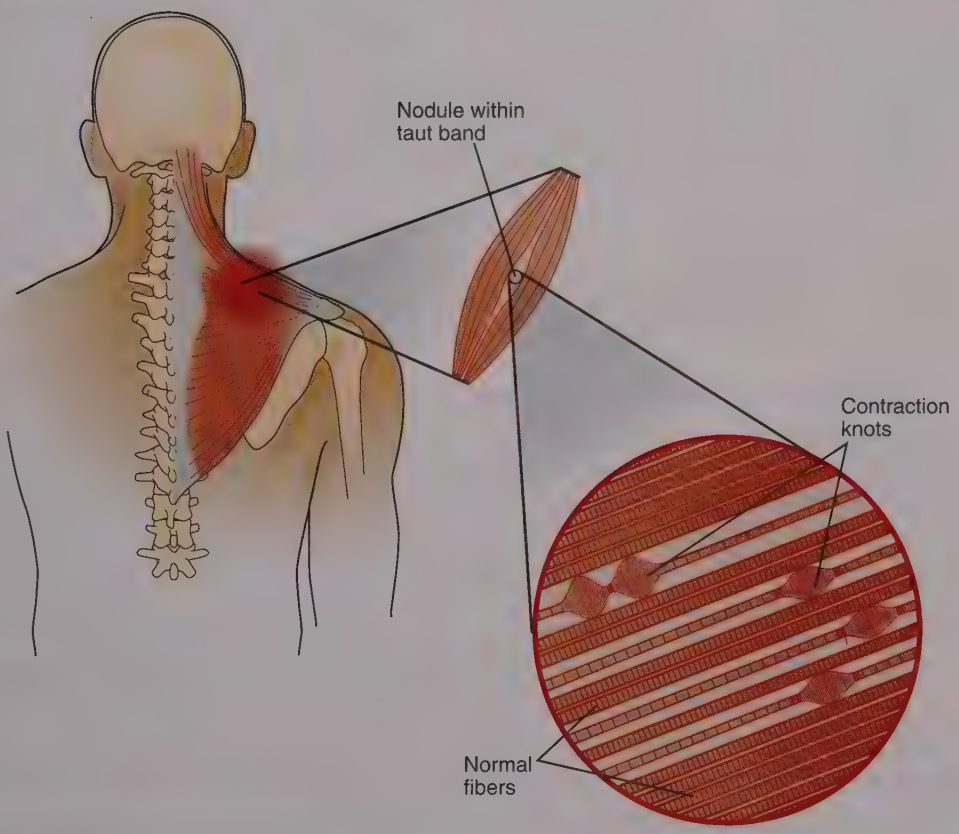


FIGURE 7.9
Myofascial adhesions.

a weak, inelastic matrix (inability to stretch) that decreases normal elasticity of the soft tissue, resulting in altered length-tension relationships (leading to altered reciprocal inhibition), altered force-couple relationships (leading to synergistic dominance), and arthrokinetic dysfunction (leading to altered joint motion) (1,5,28). Left untreated, these adhesions can begin to form permanent structural changes in the soft tissue that is evident by Davis's law.

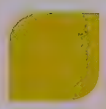
Davis's law states that soft tissue models along the lines of stress (1,5,29). Soft tissue is remodeled (or rebuilt) with an inelastic collagen matrix that forms in a random

Davis's law States that soft tissue models along the lines of stress.

fashion, meaning it usually does not run in the same direction as the muscle fibers. If the muscle fibers are lengthened, these inelastic connective tissue fibers act as roadblocks, preventing the muscle fibers from moving properly, which creates alterations in normal tissue extensibility and causes relative flexibility (9).

If a muscle is in a constant shortened state (such as the hip flexor musculature when sitting for prolonged periods every day), it will demonstrate poor neuromuscular efficiency (as a result of altered length-tension and force-couple relationships). In turn, this will affect joint motion (ankle, knee, hip, and lumbar spine) and alter movement patterns (leading to synergistic dominance). An inelastic collagen matrix will form along the same lines of stress created by the altered muscle movements. Because the muscle is consistently short and moves in a pattern different from its intended function, the newly formed inelastic connective tissue forms along this altered pattern, reducing the ability of the muscle to extend and move in its proper manner. This is why it is imperative that an integrated flexibility training program be used to restore the normal extensibility of the entire soft tissue complex (30,31).

It is essential for personal trainers to address their clients' muscular imbalances through an integrated fitness assessment and flexibility training program. If personal trainers neglect these phases of programming and simply move their clients right into a resistance or cardiorespiratory training program, this will add additional stresses to joints and muscles because they have improper mechanics and faulty recruitment patterns.



SUMMARY

The benefits of flexibility training include improving muscle imbalances, increasing joint range of motion and muscle extensibility, relieving excessive tension of muscles and joint stress, and improving neuromuscular efficiency and function. People who train in a repetitive fashion (or have jobs that require moving their bodies in repetitive ways) are at risk for pattern overload, which places stress on the body and can result in injury. Poor posture and repetitive movements may create dysfunctions in connective tissue, initiating the cumulative injury cycle. Tissue trauma creates inflammation, which leads to microspasms and decreases normal elasticity of the soft tissue.

Soft tissue rebuilds itself in a random fashion with an inelastic collagen matrix that usually does not run in the same direction as the muscle fibers. If the muscle fibers are lengthened, these inelastic connective tissue fibers act as roadblocks, creating alterations in normal tissue extensibility and causing relative flexibility. It is essential for fitness professionals to address muscular imbalances through a comprehensive fitness assessment and flexibility training program to restore the normal extensibility of the entire soft tissue complex.

The Flexibility Continuum

To better appreciate the benefits and program design considerations, personal trainers need to understand the different types of flexibility training. Flexibility, like any other form of training, should follow a systematic progression known as the flexibility continuum. There are three phases of flexibility training within the OPT model: corrective, active, and functional **Figure 7.10** (1,10,14,32,33). Moreover, it is important to note

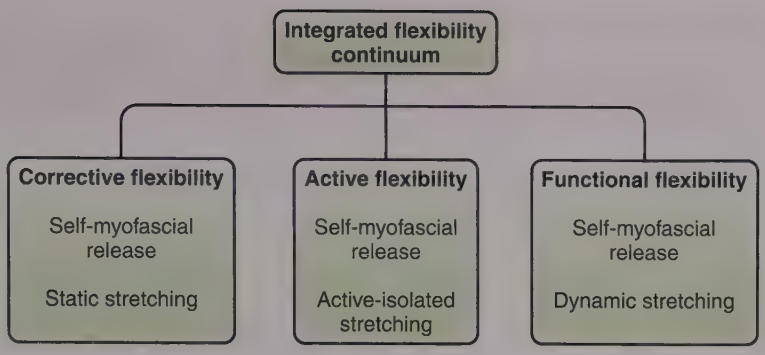


FIGURE 7.10
Integrated flexibility continuum.

that flexibility techniques should only be performed on tissues that have been identified as overactive (tight) during the assessment process.

Corrective Flexibility

Corrective flexibility is designed to increase joint ROM, improve muscle imbalances, and correct altered joint motion. Corrective flexibility includes self-myofascial release (foam roll) techniques and static stretching. Self-myofascial release uses the principle of autogenic inhibition to cause muscle relaxation, whereas static stretching can use either autogenic inhibition or reciprocal inhibition to increase muscle length depending on how the stretch is performed. Corrective flexibility is appropriate at the stabilization level (phase 1) of the OPT model.

Active Flexibility

Active flexibility uses self-myofascial release and active-isolated stretching techniques. Active-isolated stretching is designed to improve the extensibility of soft tissue and increase neuromuscular efficiency by using reciprocal inhibition. Active-isolated stretching allows for agonists and synergist muscles to move a limb through a full range of motion while the functional antagonists are being stretched (14,34,35). For example, a supine straight-leg raise uses the hip flexors and quadriceps to raise the leg and hold it unsupported, while the antagonist hamstring complex is stretched. Active flexibility would be appropriate at the strength level (phases 2, 3, and 4) of the OPT model.

Functional Flexibility

Functional flexibility uses self-myofascial release techniques and dynamic stretching. Dynamic stretching requires integrated, multiplanar soft tissue extensibility, with optimal neuromuscular control, through the full range of motion, or essentially movement without compensations (14). Therefore, if clients are compensating when performing dynamic stretches during training, then they need to be regressed to active or corrective flexibility. This form of flexibility would be appropriate at the power level (phase 5) of the OPT model or before athletic competition. It is important to remember that all functional movements occur in all three planes of motion and that injuries most often occur in the transverse plane. If the appropriate soft tissue is not extensible through the full range of movement, the risk of injury dramatically increases (2,36). Exercises that increase multiplanar soft tissue extensibility and have high levels of neuromuscular demand are preferred.



STRETCH Your Knowledge

Which Type of Stretching Should Be Used? All of Them—in a Progressive, Integrated Fashion

The best flexibility training programs incorporate multiple forms of stretching to meet the needs of the client. Each type of stretch creates different effects on the neuromuscular system, and can be used in combination to safely and effectively increase flexibility. The type of exercise will be based on the findings of the evaluation, the goals of the program, and when the stretches will be used (before exercise, after exercise).

Self-Myofascial Release (SMR)

Used to help correct existing muscle imbalances, reduce trigger points (knots within muscle) and inhibit overactive musculature. Can be used before exercise as well as after exercise.

- Hou et al. (1) in a randomized controlled trial of 119 patients found that ischemic compression therapy provides immediate pain relief and trigger point sensitivity suppression.
- Hanten et al. (2) in a randomized trial of 40 adults, found a home program, consisting of ischemic pressure and sustained stretching, was shown to be effective in reducing trigger point sensitivity and pain intensity in individuals with neck and upper back pain.

Static Stretching

Used to correct existing muscle imbalances and lengthen overactive (tight) musculature. Can be used before exercise as well as after exercise. The evidence is strong that static stretching performed daily causes increased flexibility in the stretched muscle.

- Decoster et al. (3), in a systematic review of 28 randomized control trials with 1,338 subjects, found moderately strong evidence that static stretching can increase the flexibility of the hamstring muscles, with varying duration and frequency of stretch.
- Ford et al. (4), in a randomized control trial with 35 subjects with decreased hamstring flexibility, completed static hamstring stretches with durations between 30 and 120 seconds (a control group did no stretching). All four stretching groups increased hamstring flexibility after 5 weeks, and the 30-second hold was just as effective as the 120-second hold.
- Reid and McNair (5), in a randomized controlled trial of 43 males performing 30-second static hamstring stretches for 6 weeks, found a mean gain of 10 degrees in hamstring flexibility compared with the control group.

(Continued)

- Porter et al. (6) in a randomized controlled trial with 94 subjects demonstrated that both sustained (3 minutes, 3 times daily) and intermittent stretching (5 sets of 20-second holds, 2 times daily) for the Achilles' tendon increased flexibility and decreased pain.

Active Stretching

Used to increase the extensibility of soft tissues through reciprocal inhibition. There is moderate evidence that forms of active stretching can increase joint range of motion and muscle flexibility.

- Maddig and Harmer (7) in a randomized controlled trial with 30 recreational athletes found that active-isolated stretching is effective for increasing hamstring range of motion.
- Guroian et al. (8) in a randomized trial found that static stretching and active-isolated stretching were both effective for increased range of motion and flexibility in older adults.

Dynamic Stretching

Dynamic or functional stretching is used to increase flexibility with optimal neuromuscular control. This should be used once clients have demonstrated adequate control over motions to prevent injury.

- Sherry and Best (9) in a prospective randomized comparison study of two rehabilitation programs with 24 athletes demonstrated improved functional outcomes with patients using dynamic functional movements (progressive agility and trunk stabilization) that require stabilization, proprioception, and muscle lengthening to occur simultaneously.

REFERENCES

1. Hou C-R, Tsai L-C, Cheng K-F, Chung K-C, Hong C-Z. Immediate effects of various therapeutic modalities on cervical myofascial pain and trigger-point sensitivity. *Arch Phys Med Rehabil.* 2002;83:1406-1414.
2. Hanten WP, Olson SL, Butts NL, Nowicki AL. Effectiveness of a home program of ischemic pressure followed by sustained stretch for treatment of myofascial trigger points. *Phys Ther.* 2000;80:997-1003.
3. Decoster LC, Cleland J, Altieri C, Russell P. The effects of hamstring stretching on range of motion: a systematic literature review. *J Orthop Sports Phys Ther.* 2005:35(6):377-387.
4. Ford GS, Mazzone MA, Taylor K. The effect of 4 different durations of static hamstring stretching on passive knee-extension range of motion. *J Sport Rehabil.* 2005;14(2):95-107.
5. Reid DA, McNair PJ. Passive force, angle, and stiffness changes after stretching of hamstring muscles. *Med Sci Sports Exerc.* 2004:36(11):1944-1948.
6. Porter D, Barrill E, Oneacre K, May BD. The effects of duration and frequency of Achilles tendon stretching on dorsiflexion and outcome in painful heel syndrome. *Foot Ankle Int.* 2002;23(7):619-624.
7. Maddig TR, Harmer P. Active-isolated stretching is not more effective than static stretching for increased hamstring ROM. *Med Sci Sports Exerc.* 2002;34(5) p S151.
8. Guroian L, Walsh C, Diaz L, Marra A, Wygand J, Otto R. The effects of active isolated stretching on flexibility and function in older adults. *Med Sci Sports Exerc.* 2008;40(5):S373.
9. Sherry MA, Best TM. A comparison of 2 rehabilitation programs in the treatment of acute hamstring strains. *J Orthop Sports Phys Ther.* 2004;34(3):116-125.

SUMMARY

Flexibility training should be progressive, systematic, and based on an assessment. There are three phases of flexibility training: corrective, active, and functional. Corrective flexibility improves muscle imbalances and altered joint motion by using self-myofascial release and static stretching. Active flexibility improves the extensibility of soft tissue and increases neuromuscular efficiency by using self-myofascial release and active-isolated stretching. Functional flexibility improves the extensibility of soft tissue and increases neuromuscular efficiency by using self-myofascial release and dynamic stretches that incorporate multiplanar movements through the full range of motion.

Stretching Techniques

As previously discussed, proper stretching enhances flexibility and can be viewed as a continuum. The flexibility continuum consists of specific forms of stretching. For example, corrective flexibility uses self-myofascial release and static stretching; active flexibility uses self-myofascial release and active-isolated stretching; and functional flexibility uses self-myofascial release and dynamic stretching Table 7.2. Each form of stretching manipulates the receptors of the nervous system, which in turn allows for alteration of muscle extensibility.

TABLE 7.2 Examples of Stretching within the Flexibility Continuum

© ilolab/ShutterStock, Inc.

Flexibility Type	Type of Stretching	Examples
Corrective flexibility	Self-myofascial release (SMR)	SMR: gastrocnemius/soleus SMR: adductors SMR: latissimus dorsi
	Static stretching	Static gastrocnemius stretch Static adductor stretch Static latissimus dorsi stretch
Active flexibility	Self-myofascial release (SMR)	SMR: adductors SMR: latissimus dorsi SMR: thoracic spine
	Active-isolated stretching	Active standing adductor stretch Active latissimus dorsi ball stretch Active pectoral wall stretch
Functional flexibility	Self-myofascial release (SMR)	SMR: gastrocnemius/soleus SMR: TFL/IT band SMR: latissimus dorsi
	Dynamic stretching	Prisoner squat Multiplanar lunge Tube walking: side to side Medicine ball lift and chop

TFL, tensor fascia latae; IT, iliotibial.

©nolan/ShutterStock, Inc.

Myofascial Release

Self-myofascial release is a stretching technique that focuses on the neural system and fascial system in the body (or the fibrous tissue that surrounds and separates muscle tissue). By applying gentle force to an adhesion or "knot," the elastic muscle fibers are altered from a bundled position (which causes the adhesion) into a straighter alignment with the direction of the muscle or fascia. The gentle pressure (applied with implements such as a foam roll) will stimulate the Golgi tendon organ and create autogenic inhibition, decreasing muscle spindle excitation and releasing the hypertonicity (tension) of the underlying musculature. In other words, gentle pressure (similar to a massage) breaks up knots within the muscle and helps to release unwanted muscular tension.

It is crucial to note that when a person is using self-myofascial release he or she must find a tender spot (which indicates the presence of muscle hypertonicity) and sustain pressure on that spot for a minimum of 30 seconds. This will increase the Golgi tendon organ activity and decrease muscle spindle activity, thus triggering the autogenic inhibition response. It may take longer, depending on the client's ability to consciously relax. This process will help restore the body back to its optimal level of function by resetting the proprioceptive mechanisms of the soft tissue (37). Self-myofascial release is suggested before stretching because breaking up fascial adhesions (knots) may potentially improve the tissue's ability to lengthen through stretching techniques. In addition, it can be used during the cool-down process.

STRETCHES FOR MYOFASCIAL RELEASE

©nolan/ShutterStock, Inc.

Gastrocnemius/Soleus (Calves)



Preparation

1. Place foam roll under mid-calf.
2. Cross left leg over right leg to increase pressure (optional).

Movement

3. Slowly roll calf area to find the most tender spot.
4. Once identified, hold tender spot until the discomfort is reduced (minimum 30 seconds).

Tensor Fascia Latae (TFL)/Iliotibial (IT) Band



Preparation

1. Lie on one side, the foam roll just in front of the hip. Cross the top leg over lower leg, with foot touching the floor.

Movement

2. Slowly roll from hip joint to lateral knee to find the most tender spot.
3. Once identified, hold tender spot until the discomfort is reduced (minimum 30 seconds).

STRETCHES FOR MYOFASCIAL RELEASE continued

Adductors



Preparation

1. Lie prone with one thigh flexed and abducted and the foam roll in the groin region, inside the upper thigh.

Movement

2. Slowly roll the inner thigh area to find the most tender spot.
3. Once identified, hold tender spot until the discomfort is reduced (minimum 30 seconds).

Piriformis



Preparation

1. Sit on top of the foam roll, positioned on the back of the hip. Cross one foot to the opposite knee.

Movement

2. Lean into the hip of the crossed leg. Slowly roll on the posterior hip area to find the most tender spot.
3. Once identified, hold tender spot until the discomfort is reduced (minimum 30 seconds).

Latissimus Dorsi



Preparation

1. Lie on the floor on one side with the arm closest to the floor outstretched and thumb facing upward.
2. Place the foam roll under the arm (axillary region).

Movement

3. Slowly move back and forth to find the most tender spot.
4. Once identified, hold tender spot until the discomfort is reduced (minimum 30 seconds).

© Hotlab/ShutterStock, Inc

Static Stretching

Static stretching is the process of passively taking a muscle to the point of tension and holding the stretch for a minimum of 30 seconds (1,2,11). This is the traditional form of stretching that is most often seen in fitness today. It combines low force with longer duration (14,38).

Static stretching The process of passively taking a muscle to the point of tension and holding the stretch for a minimum of 30 seconds.

By holding the muscle in a stretched position for a prolonged period, the Golgi tendon organ is stimulated and produces an inhibitory effect on the muscle spindle (autogenic inhibition). This allows the muscle to relax and provides for better elongation of the muscle Table 7.3 (5,39). In addition, contracting the antagonistic musculature while holding the stretch can reciprocally inhibit the muscle being stretched, allowing it to relax and enhancing the stretch. For example, when performing the kneeling hip flexor stretch, an individual can contract the hip extensors (gluteus maximus) to reciprocally inhibit the hip flexors (psoas, rectus femoris), allowing for greater lengthening of these muscles. Another example would be to contract the quadriceps when performing a hamstring stretch.

Static stretching should be used to decrease the muscle spindle activity of a tight muscle before and after activity. Detailed explanations of various static stretching techniques are described below.

TABLE 7.3 **Static Stretching Summary**

© Hotlab/ShutterStock, Inc.

Type of Stretch	Mechanism of Action	Acute Variables	Examples
Static stretch	Autogenic inhibition or reciprocal inhibition (depending how stretch is performed)	1–3 sets Hold each stretch 30 seconds	• Gastrocnemius stretch • Kneeling hip flexor stretch • Standing adductor stretch • Pectoral wall stretch



STRETCH Your Knowledge

Does Stretching Decrease Strength?

Recently, researchers have suggested that static stretching, performed immediately before exercise or athletic events, may decrease muscular strength and impair performance. The research in this area is still developing, but there are several studies that have found that static stretching may decrease maximal strength and power for up to 10 minutes (1–5). Naturally, this has caused some controversy for exercise and health professionals. Should patients and clients stop stretching before exercise? Is stretching "bad" for athletic performance?

The best answer is both "yes" and "no." There is mounting scientific evidence that static stretching may decrease strength and power. This appears to be limited to just static stretching, often performed in an acute fashion. However, research has also demonstrated increases in vertical jump, muscular strength, power, and balance ability after a regular (chronic) stretching program (6–11).

(Continued)

In addition, there is no conclusive evidence that patients and clients with known motion limitations and flexibility imbalances have the same reaction to static stretching. Finally, there is little evidence that stretching influences *overall* athletic performance (for example, although quadriceps maximal strength may be decreased, there is no evidence that this prevents a soccer player from kicking the ball with maximal velocity). Therefore, stretching before physical activity may still be beneficial. The current recommendations for the use of stretching before exercise are as follows:

- Acute static stretching, held for more than 30 seconds, may decrease strength and power. Therefore, athletes and others who will be engaging in *maximal effort, explosive* activities (such as high jump, sprinting, or power lifting) may not want to perform static stretching before the event *unless* muscle imbalances are present. If static stretching is used, care should be taken so that only the targeted (tight, overactive) muscles are stretched, and this should be followed by active-isolated or dynamic stretches to increase motorneuron excitability.
- Static stretching may be used to correct muscle imbalances and increase joint range of motion before activity in most patients and clients, as part of a progressive, integrated program.
- Active and dynamic stretching may be used without risking a loss of strength or power. These two forms of stretching may be the most appropriate before physical activity in clients with no identified muscle imbalances.

REFERENCES

1. Fowles JR, Sale DG, MacDougall JD. Reduced strength after passive stretch of the human plantarflexors. *J Appl Physiol.* 2000;89:1179-1188.
2. Knudson D, Noffal G. Time course of stretch-induced isometric strength deficits. *Eur J Appl Physiol.* 2005;94:348-351.
3. Kokkonen J, Nelson AG, Cornwell A. Acute muscle stretching inhibits maximal strength performance. *Res Q Exerc Sport.* 1998;69:411-415.
4. Marek SM, Cramer JT, Fincher AL, et al. Acute effects of static and proprioceptive neuromuscular facilitation stretching on muscle strength and power output. *J Athl Train.* 2005;40:94-103.
5. Power K, Behm D, Cahill F, Carroll M, Young W. An acute bout of static stretching: effects on force and jumping performance. *Med Sci Sports Exerc.* 2004;36:1389-1396.
6. Shrier I. Does stretching improve performance? A systematic and critical review of the literature. *Clin J Sport Med.* 2004;14(5):267-273.
7. Hunter JP, Marshall RN. Effects of power and flexibility training on vertical jump technique. *Med Sci Sports Exerc.* 2002;34(3):478-486.
8. Gajdosik RL, Vander Linden DW, McNair PJ, Williams AK, Riggin TJ. Effects of an eight-week stretching program on the passive-elastic properties and function of the calf muscles of older women. *Clin Biomech (Bristol, Avon).* 2005;20(9):973-983.
9. Kokkonen J, Nelson AG, Eldredge C, Winchester JB. Chronic static stretching improves exercise performance. *Med Sci Sports Exerc.* 2007;39(10):1825-1831.
10. Wilson GJ, Elliott BC, Wood GA. Stretch shorten cycle performance enhancement through flexibility training. *Med Sci Sports Exerc.* 1992;24(1):116-123.
11. LaRoche DP, Lussier MV, Roy SJ. Chronic stretching and voluntary muscle force. *J Strength Cond Res.* 2008;22(2):589-596.
12. Bazett-Jones DM, Gibson MH, McBride JM. Sprint and vertical jump performances are not affected by six weeks of static hamstring stretching. *J Strength Cond Res.* 2008;22(1):25-33.

© iHotab/ShutterStock, Inc

STATIC STRETCHES

© iHotab/ShutterStock, Inc

Static Gastrocnemius Stretch



Preparation

1. Stand facing a wall or stable object.
2. Extend one leg back, keeping the knee and foot straight and the back heel on the floor.

Movement

3. Draw navel inward.
4. Keep rear foot flat, with foot pointed straight ahead. Do not allow the rear foot to pronate.
5. Bend arms and lean forward toward the wall. Keep the gluteal muscles and quadriceps tight and the heel on the ground.
6. Hold for 30 seconds.

TECHNIQUE

Make sure the gluteal muscles and quadriceps are activated to keep the knee in full extension. This will enhance the stretch to the gastrocnemius.

Static Standing TFL Stretch



Preparation

1. Stand in a staggered stance with the front leg slightly bent and rear leg straight.
2. Externally rotate back leg.

Movement

3. Draw navel inward.
4. Squeeze gluteal muscles while rotating pelvis posteriorly.
5. Slowly move body forward until a mild tension is achieved in the front of the hip being stretched.
6. As a progression, raise the arm (on the same side as the back leg) up and over to the opposite side while maintaining pelvis position.
7. Hold side bend position and slowly rotate posteriorly as illustrated.
8. Hold for 30 seconds.
9. Switch sides and repeat.

TECHNIQUE

Make sure the gluteal musculature is contracted during the stretch. This will help reciprocally inhibit the TFL, allowing for greater lengthening of the TFL.

STATIC STRETCHES *continued*

© iStock/ShutterStock, Inc.

Static Kneeling Hip Flexor Stretch



Preparation

1. Kneel with front and back legs bent at a 90-degree angle.
2. Internally rotate back hip to target the psoas musculature or maintain a neutral position to target the rectus femoris.

Movement

3. Draw navel inward.
4. Squeeze gluteal muscles of the side being stretched while rotating pelvis posteriorly.
5. Slowly move body forward until a mild tension is achieved in the front of the hip being stretched.
6. As a progression, raise arm, side bend to opposite side, and rotate posteriorly as illustrated.
7. Hold for 30 seconds.

TECHNIQUE Placing a foam pad underneath the knee (not illustrated) may improve comfort for the client, especially if stretching on a hard surface.

© Notab/ShutterStock, Inc.

STATIC STRETCHES *continued*

Static Standing Adductor Stretch



TECHNIQUE

Be sure to take a wider stance than shoulders-width apart to ensure optimal lengthening. This stretch can also be performed from a kneeling position or seated on a stability ball to reduce demand caused by maintaining a static lunge position.

Preparation

1. Stand in a straddled stance with the feet beyond shoulders-width apart. Extend one leg back until the toe of the back leg is in line with the heel of the other foot. Both feet should be pointed straight ahead.

Movement

2. Draw navel inward and posteriorly rotate the pelvis.
3. Slowly move in a sideways motion (side lunge) until a stretch in the straight leg's groin area is felt.
4. Hold for 30 seconds.

Static Latissimus Dorsi Ball Stretch



Preparation

1. Kneel in front of a stability ball.
2. Place one arm on ball, with thumb pointed straight up in the air.

Movement

3. Draw navel upward.
4. Posteriorly rotate the pelvis.
5. Slowly reach the arm straight out by rolling the ball forward.
6. Hold for 30 seconds.

TECHNIQUE

If this stretch causes any pinching in the shoulder, perform the stretch with the palm down on the ball. To increase the stretch, slightly adduct the outstretched arm across the body.

STATIC STRETCHES *continued*

Static Pectoral Stretch



Preparation

1. Stand against an object and form a 90-90-degree angle with your arm.

Movement

2. Draw your navel inward.
3. Slowly lean forward until a slight stretch is felt in the anterior shoulder and chest region.
4. Hold the stretch for 30 seconds.

TECHNIQUE

Make sure the shoulders do not elevate during the stretch. This is an example of relative flexibility and decreases the effectiveness of the stretch.

Static Upper Trapezius/Scalene Stretch



Preparation

1. Stand in optimal posture.

Movement

2. Draw navel inward.
3. Retract and depress the scapula on the side being stretched.
4. Tuck chin and slowly laterally flex the head, pulling one ear toward the same side shoulder.
5. Hold the stretch position for 30 seconds.
6. Switch sides and repeat.

TECHNIQUE

As with the pectoral stretch, keep the shoulder of the side being stretched down and retract it by depressing the scapula on the side being stretched. To help accomplish this, one can place the arm on the side being stretched behind the body.

© Ilolab/ShutterStock, Inc

Active-Isolated Stretching

Active-isolated stretching is the process of using agonists and synergists to dynamically move the joint into a range of motion (14,33). This form of stretching increases motorneuron excitability, creating reciprocal inhibition of the muscle being stretched. The active supine biceps femoris stretch is a good example of active-isolated stretching (1,14). The quadriceps extends the knee. This enhances the stretch of the biceps femoris in two ways. First, it increases the length of the biceps femoris. Second, the contraction of the quadriceps causes reciprocal inhibition (decreased neural drive and muscle spindle excitation) of the hamstring complex, which allows it to elongate.

Active-isolated stretch The process of using agonists and synergists to dynamically move the joint into a range of motion.

Active-isolated stretches are suggested for preactivity warm-up (such as before sports competition or high-intensity exercise), as long as no postural distortion patterns are present. If an individual possesses muscle imbalances, active-isolated stretching should be performed after self-myofascial release and static stretching for muscles determined as tight or overactive during the assessment process. Typically, 5 to 10 repetitions of each stretch are performed and held for 1 to 2 seconds each. Detailed explanations of various active-isolated stretches are given below Table 7.4.

TABLE 7.4 Active-Isolated Stretching Summary

© Ilolab/ShutterStock, Inc.

Type of Stretch	Mechanism of Action	Acute Variables	Examples
Active-isolated stretch	Reciprocal inhibition	1–2 sets Hold each stretch 1–2 seconds for 5–10 repetitions	• Active supine biceps femoris stretch • Active kneeling quadriceps stretch • Active standing adductor stretch • Active pectoral wall stretch

ACTIVE-ISOLATED STRETCHES

© nolah/ShutterStock, Inc.

Active Gastrocnemius Stretch with Pronation and Supination



TECHNIQUE

Make sure when performing the stretch that the majority of the motion is coming from internal and external rotation of the hip, which in turn causes rotation at the knee and eversion and inversion on the foot and ankle.

Preparation

1. Stand near a wall or sturdy object.
2. Bring one leg forward for support. Use upper body and lean against object.

Movement

3. Draw navel inward.
4. Keep rear foot on the ground, with opposite hip flexed.
5. Slowly move through hips, creating controlled supination and pronation through the lower extremity.
6. Hold for 1–2 seconds and repeat for 5–10 repetitions.

Active Supine Biceps Femoris Stretch



Preparation

1. Lie supine on floor with legs flat.
2. Flex, adduct, and slightly internally rotate the hip of the side being stretched while keeping the knee flexed.
3. Place the opposite hand behind the knee of the leg being stretched.

Movement

4. Draw navel inward.
5. With hand supporting leg, extend the knee.
6. Hold for 1–2 seconds and repeat for 5–10 repetitions.

TECHNIQUE

Adducting and internally rotating the hip places more of an emphasis on the short head of the biceps femoris.

ACTIVE-ISOLATED STRETCHES *continued*

iStock/ShutterStock, Inc.

Active Standing TFL Stretch



TECHNIQUE

As with the static stretch, make sure the gluteal muscles are contracted when going into the stretch. This will help in enhancing neuromuscular efficiency between the hip flexors and hip extensors.

Preparation

1. Stand in a staggered stance with the front leg slightly bent and rear leg straight.
2. Externally rotate back leg.

Movement

3. Draw navel inward and raise arm overhead.
4. Squeeze gluteal muscles while rotating pelvis posteriorly.
5. Stride forward, in a controlled manner, until a mild tension is achieved in the front of the hip being stretched. Side bend and rotate posteriorly as illustrated.
6. Hold for 1–2 seconds and repeat for 5–10 repetitions.

Active Kneeling Hip Flexor Stretch



Preparation

1. Kneel with front and back legs bent at a 90-degree angle.
2. Internally rotate back hip to target the psoas musculature or maintain a neutral position to target the rectus femoris.

Movement

3. Draw navel inward and raise arm overhead.
4. Squeeze gluteal muscles of the side being stretched while rotating pelvis posteriorly.
5. Slowly move body forward until a mild tension is achieved in the front of the hip being stretched. Side bend and rotate posteriorly.
6. Hold for 1–2 seconds and repeat for 5–10 repetitions.

TECHNIQUE

Internally rotating the back hip targets the psoas because this muscle concentrically performs hip flexion and external rotation.

ACTIVE-ISOLATED STRETCHES *continued*

© Istvan/ShutterStock, Inc.

Active Standing Adductor Stretch



TECHNIQUE

Make sure to keep the hips level when going into the stretch.

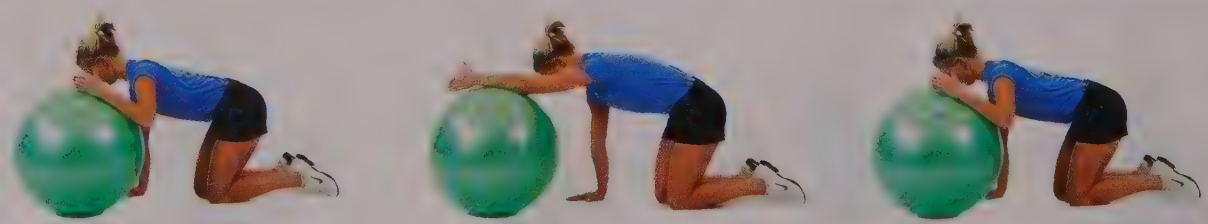
Preparation

1. Stand in a straddled stance with the feet more than shoulders-width apart. Extend one leg back until the toe of the back leg is in line with the heel of the other foot. Both feet should be pointed straight ahead.

Movement

2. Draw navel inward and posteriorly rotate the pelvis.
3. Slowly move in a sideways motion (side lunge) until a stretch in the straight leg's groin area is felt.
4. Hold for 1–2 seconds and repeat for 5–10 repetitions.

Active Latissimus Dorsi Ball Stretch



Preparation

1. Kneel in front of stability ball.
2. Place one arm on ball with thumb straight up in the air.

Movement

3. Draw navel upward.
4. Maintaining core control, roll ball out until a comfortable stretch is felt.
5. Hold stretch for 1–2 seconds and repeat for 5–10 repetitions.
6. Switch sides and repeat.

TECHNIQUE

Make sure to initiate the stretch by going into a posterior pelvic tilt. This will take the origin and insertion of the latissimus dorsi further apart, enhancing the stretch.

ACTIVE-ISOLATED STRETCHES continued

Active Pectoral Stretch



Preparation

1. Stand against an object and form a 90-90-degree angle with the arm as depicted.

Movement

2. Draw navel inward.
3. Slowly lean forward until a slight stretch is felt in the anterior shoulder and chest region.
4. Hold stretch for 1–2 seconds and repeat for 5–10 repetitions.

TECHNIQUE

Retract the scapulae when going into the stretch. This will reciprocally inhibit the pectoralis major and minor, enhancing the stretch.

Active Upper Trapezius/Scalene Stretch



Preparation

1. Stand with optimal posture.

Movement

2. Draw navel inward.
3. Tuck chin and slowly laterally flex the head, pulling one ear toward the same-side shoulder while retracting and depressing the same-side shoulder complex.
4. Hold stretch for 1–2 seconds and repeat for 5–10 repetitions.

TECHNIQUE

If tingling is felt down the arm and into the finger, decrease the range of motion of the stretch. This will take stress off of the nerve. Also make sure the head stays in a neutral position during the stretch. Do not allow the head to jut forward.

Dynamic Stretching

Dynamic stretch The active extension of a muscle, using force production and momentum, to move the joint through the full available range of motion.

Dynamic stretching uses the force production of a muscle and the body's momentum to take a joint through the full available range of motion Table 7.5. Dynamic stretching uses the concept of reciprocal inhibition to improve soft tissue extensibility. One can perform one set of 10 repetitions using 3 to 10 dynamic stretches. Hip swings, medicine ball rotations, and walking lunges are good examples of dynamic stretching (1,14). Dynamic stretching is also suggested as a warm-up before athletic activity, as long as no postural distortion patterns are present. If an individual does possess muscle imbalances, self-myofascial release and static stretching should precede dynamic stretching for overactive or tight muscles identified during the assessment process. It is recommended that the client have good levels of tissue extensibility, core stability, and balance capabilities before undertaking an aggressive dynamic stretching program.

TABLE 7.5 Dynamic Stretching Summary

© Rolab/ShutterStock, Inc.

Type of Stretch	Mechanism of Action	Acute Variables	Examples
Dynamic stretch	Reciprocal inhibition	1–2 sets	• Prisoner squats
		10–15 repetitions	• Multiplanar lunges
		3–10 exercises	• Single-leg squat touchdowns
			• Tube walking
			• Medicine ball lift and chop

DYNAMIC STRETCHES

© Rolab/ShutterStock, Inc.

Prisoner Squat



Preparation

1. Stand in proper alignment, with the hands behind the head.

Movement

2. Draw navel inward.
3. Lower to a squat position, under control and without compensation (toes straight ahead, knees in line with the toes).
4. Extend hips, knees, and ankles and repeat.
5. Perform 10 repetitions.

TECHNIQUE

As a progression add a calf raise after extending the hips, knees and ankles.

DYNAMIC STRETCHES *continued*

Multiplanar Lunge with Reach



Preparation

1. Stand in proper alignment with hands on hips and feet straight ahead.

Movement

2. Draw navel inward.
3. While maintaining total body alignment, step forward (sagittal plane), and descend into a lunge position while reaching forward as illustrated.
4. Use hip and thigh muscles to push up and back to the start position.
5. Perform 10 repetitions.
6. Repeat on opposite leg.
7. Progress to side lunges with reach (frontal plane), followed by turning lunges with reach (transverse plane).

Single-Leg Squat Touchdown



Preparation

1. Stand on one leg in optimal posture, keeping raised leg parallel to the standing leg. Imagine the floating leg is resting on an imaginary phonebook.

Movement

2. Draw navel inward.
3. Squat, in a controlled manner, bending the ankle, knee, and hip while reaching the opposite hand near the standing leg toe.
4. While maintaining drawn-in maneuver and gluteal activity, return to starting position.
5. Perform 10 repetitions.
6. Repeat on opposite side.

TECHNIQUE

Make sure the knee is tracking in line with the second and third toes. Do not allow the knee of the squatting leg to move inside the foot.

DYNAMIC STRETCHES *continued*

Tube Walking: Side-to-Side



Preparation

1. Stand with feet hip-width apart, knees slightly bent, and feet straight ahead.
2. Place tubing around lower leg.

Movement

3. Draw navel inward.
4. Keep feet straight ahead and take 10 small steps sideways, without allowing knees to cave inward.
5. Repeat in the opposite direction.

TECHNIQUE

Make sure the toes stay straight ahead and do not turn the feet out when stepping. This exercise is very effective for improving gluteus medius and core activation.

Medicine Ball Lift and Chop



Preparation

1. Stand with feet hip-width apart, knees slightly bent and feet straight ahead.
2. Grasp a medicine ball with both hands and keep elbows fully extended.

Movement

3. Draw belly button inward.
4. Starting from optimal posture, initiate the rotational movement from the trunk outward, lifting the medicine ball from a low position to a high position.
5. Allow the hips to pivot on the back foot as the motion nears end range.
6. Perform 10 repetitions.
7. Repeat on opposite side.

TECHNIQUE

Allow for the hips to rotate during both portions of this exercise (chop and lift). This will improve arthrokinematics through the lumbo-pelvic-hip complex.

SUMMARY

Each type of flexibility training consists of specific stretching techniques. Corrective flexibility uses self-myofascial release and static stretching; active flexibility uses self-myofascial release and active-isolated stretching; and functional flexibility uses self-myofascial release and dynamic stretching. Self-myofascial release applies gentle pressure on muscle hypertonicity for a minimum of 30 seconds. The force applied creates autogenic inhibition, decreasing muscle spindle excitation and releasing the muscle hypertonicity. These techniques are suggested before stretching, as well as for cool-down. Static stretching passively takes a muscle to the point of tension and holds it there for a minimum of 30 seconds. Static stretches can be used before activity to decrease overactivity of tight muscles and to increase range of motion. However, care should be used when static stretching is used before certain activities that involve explosive movements or maximal strength. Static stretches should also be used after activity to reset muscles back to their optimal resting lengths. Active-isolated stretches use agonists and synergists to dynamically move joints into their ranges of motion. These are suggested for preactivity warm-up (5 to 10 repetitions, held for 1–2 seconds each). Dynamic stretches use force production and momentum to take a joint through the full available range of motion. These are suggested for preactivity warm-up as well.

Controversial Stretches

Although stretching and flexibility training have numerous benefits, performing certain types of stretches may cause injury. Flexibility training, like any other physical activity, carries varying degrees of injury risk. Although most stretches (when performed correctly with proper posture and technique) are very safe, there are a few controversial stretches that may be potentially dangerous:

1. Inverted hurdler's stretch **Figure 7.11**: This stretch is believed to place high stress on the inside of the knee (medial collateral ligament) and may cause pain and stress on the kneecap (patella). This stretch should not be performed by anyone with a history of knee or low-back pain, and most health professionals believe this stretch should not be performed by most patients or clients.



FIGURE 7.11

Inverted hurdler's stretch.

2. Plow **Figure 7.12**: A common posture from yoga. Because of the inverted nature of this stretch (head is lower than the hips), this stretch places high stress on the neck and spine. If this stretch is not done with exact technique, it may place the spine at risk of injury. Patients or clients with a history of neck or back injury should not perform this stretch owing to the high stress it places on these structures. This position should also be avoided by individuals with high blood pressure (hypertension). In addition, many practitioners of yoga will discourage menstruating females from performing inverted stretches (although there is no research to suggest there are negative effects on the body if menstruating females perform these movements).
3. Shoulder stand **Figure 7.13**: Another common posture from yoga, and another inverted stretch. As with the plow, this position places high stress on the neck, shoulders, and spine. It should be avoided in patients with hypertension or any history of neck or spine injury, or if exact technique is not used.
4. Straight-leg toe touch **Figure 7.14**: One of the most common stretches for the hamstring complex. This position may place the vertebrae and the cartilage discs in the low back under high stress. Any client or patient with a history of herniated discs or nerve pain that runs in the back of the leg should avoid this stretch. In addition, clients with poor flexibility may attempt to hyperextend the knees during this stretch, which may place high stress on the ligaments of the knee.
5. Arching quadriceps **Figure 7.15**: This stretch is designed to stretch the quadriceps and hip flexors. This position places very high stress on the kneecap and the

FIGURE 7.12

Plow.



FIGURE 7.13

Shoulder stand.





FIGURE 7.14
Straight-leg toe touch.



FIGURE 7.15
Arching quadriceps.

other tissues on the front of the knee joint. Any client or patient with a history of knee injury should avoid this stretch. Owing to the high stress (compression) of the kneecap into the knee during this stretch (which may cause damage to the cartilage), most health-care professionals discourage anyone from performing this exercise.

So, why would anyone perform these stretches if they are dangerous? Some of these positions are required in certain sports or activities (for instance, the inverted hurdler's stretch mimics the position of a hurdler going over a hurdle). Others are traditional positions used in martial arts, gymnastics, or yoga. However, for most clients, there are safer positions that can be used to stretch the targeted muscles. Therefore, all clients should be properly educated about stretching technique and posture, and the safest exercises should be used to meet the goals of the exercise program.

Practical Application of Flexibility Training for Movement Compensations

As previously mentioned, there are some common movement compensations that must be addressed to ensure the safety and effectiveness of a training program. Proper flexibility is the first step to addressing these problems. Table 7.6 provides the common compensations seen during the assessment process, associated overactive and underactive muscles, and corrective strategies for each (flexibility and strengthening exercises). Further chapters in the text will provide proper instruction for the sample strengthening exercises that can be implemented to help strengthen the underactive muscles in each of the compensations.

TABLE 7.5 Compensations, Muscle Imbalances, and Corrective Strategies

View	Check-point	Compensation	Probable Overactive Muscles	Probable Underactive Muscles	Sample SMR (foam roll) and Static Stretch Techniques	Sample Strengthening Exercises
Anterior	Feet	Turn out	Soleus Lateral gastrocnemius Biceps femoris (short head)	Medial gastrocnemius Medial hamstring complex Gracilis Sartorius Popliteus	SMR: Gastrocnemius/soleus SMR: Biceps femoris (short head) Static gastrocnemius stretch Static supine biceps femoris stretch	Single-leg balance reach
	Knees	Move inward	Adductor complex Biceps femoris (short head) Tensor fascia latae Vastus lateralis	Gluteus medius/maximus Vastus medialis oblique (VMO)	SMR: Adductors SMR: TFL/IT band Static supine biceps femoris stretch Static standing TFL stretch	Tube walking: side to side
Lateral	LPHC	Excessive forward lean	Soleus Gastrocnemius Hip flexor complex (TFL, rectus femoris, psoas) Abdominal complex (rectus abdominis, external obliques)	Anterior tibialis Gluteus maximus Erector spinae	SMR: Gastrocnemius/soleus SMR: Quadriceps Static gastrocnemius stretch Static kneeling hip flexor stretch	Quadruped arm/ opposite leg raise Ball wall squats
		Low back arches	Hip flexor complex (TFL, rectus femoris, psoas) Erector spinae Latissimus dorsi	Gluteus maximus Hamstring complex Intrinsic core stabilizers (transverse abdominis, multifidus, transversospinalis, internal oblique, pelvic-floor muscles)	SMR: Quadriceps SMR: Latissimus dorsi Static kneeling hip flexor stretch Static latissimus dorsi ball stretch	Quadruped arm/ opposite leg raise Ball wall squats
	Upper body	Arms fall forward	Latissimus dorsi Teres major Pectoralis major/minor	Mid/lower trapezius Rhomboids Rotator cuff (supraspinatus, infraspinatus, teres minor, subscapularis)	SMR: Thoracic spine SMR: Latissimus dorsi Static latissimus dorsi ball stretch Static pectoral wall stretch	Squat to row
		Shoulders elevate (pushing/pulling assessment)	Upper trapezius Sternocleidomastoid Levator scapulae	Mid/lower trapezius	SMR: Upper trapezius (Thera Cane) Static stretch upper trapezius/ scalene stretch	Ball cobra
		Head protrudes forward (pushing/pulling assessment)	Upper trapezius Sternocleidomastoid Levator scapulae	Deep cervical flexors	SMR: Upper trapezius (Thera Cane) Static stretch upper trapezius/ scalene stretch	Chin tuck (keep head in neutral position during all exercises)

© Tatiana/Shutterstock, Inc.

Filling in the Template

After a fitness assessment, the flexibility portion of the template can now be filled in. On the template, select the form of flexibility your client requires. Go to the warm-up section and input the areas to be addressed **Figure 7.16**. For beginning clients and those requiring correction of postural imbalance, corrective flexibility (self-myofascial release and static stretching) is used before and after training sessions (as well as at home, on off days). Be sure to follow the flexibility guidelines for postural distortion patterns found in this chapter (Table 7.6). Corrective flexibility will be used during the first phase (Stabilization Endurance Training) of the OPT model.

With a proper progression through the flexibility continuum (and as the client's ability dictates), active (self-myofascial release and active-isolated stretching) and functional (self-myofascial release and dynamic stretching) flexibility can be implemented later in the strength and power levels of the OPT model. The use of flexibility techniques can be a great warm-up, but also may be used as a cool-down, especially self-myofascial release and static stretching. On the template, go to the cool-down section and, as for the warm-up, input the form of stretching to be implemented and the areas to be addressed.

Professional's Name: Brian Sutton

CLIENT'S NAME: JOHN SMITH **DATE: 5/01/13**

GOAL: FAT LOSS **PHASE: 1 STABILIZATION ENDURANCE**

WARM-UP

Exercise:	Sets	Duration	Coaching Tip
SMR: Calves, IT Band, Adductors	1	30 s.	Hold each tender area for 30 sec
Static Stretch: Calves, Hip Flexors, Adductors	1	30 s.	Hold each stretch for 30 sec

CORE / BALANCE / PLYOMETRIC

Exercise:	Sets	Reps	Tempo	Rest	Coaching Tip

SPEED, AGILITY, QUICKNESS

Exercise:	Sets	Reps	Rest	Coaching Tip

RESISTANCE

Exercise:	Sets	Reps	Tempo	Rest	Coaching Tip

COOL-DOWN

Exercise:	Sets	Duration	Coaching Tip

Coaching Tips:

National Academy of Sports Medicine

FIGURE 7.16 OPT template.

© nolah/ShutterStock, Inc.

REFERENCES

1. Alter MJ. *Science of Flexibility*. 2nd ed. Champaign, IL: Human Kinetics; 1996.
2. Bandy WD, Irion JM, Briggler M. The effect of time and frequency of static stretching on flexibility of the hamstring muscles. *Phys Ther*. 1997;77(10):1090-1096.
3. Clanton TO, Coupe KJ. Hamstring strains in athletes: diagnosis and treatment. *J Am Acad Orthop Surg*. 1998;6(4):237-248.
4. Condon SA. Soleus muscle electromyographic activity and ankle dorsiflexion range of motion during four stretching procedures. *Phys Ther*. 1987;67:24-30.
5. Chaitow L. *Muscle Energy Techniques*. New York: Churchill Livingstone; 1997.
6. Janda V. Muscle spasm—a proposed procedure for differential diagnosis. *Man Med*. 1999;6136-6139.
7. Liebenson C. Integrating Rehabilitation into chiropractic practice (blending active and passive care). In: Liebenson C, ed. *Rehabilitation of the Spine*. Baltimore: Williams & Wilkins; 1996:13-44.
8. Poterfield J, DeRosa C. *Mechanical Low Back Pain: Perspectives in Functional Anatomy*. Philadelphia: WB Saunders; 1991.
9. Gossman MR, Sahrman SA, Rose SJ. Review of length-associated changes in muscle: experimental evidence and clinical implications. *Phys Ther*. 1982;62:1799-1808.
10. Halbertsma JPK, Van Bulhuis AI, Goeken LNH. Sport stretching: effect on passive muscle stiffness of short hamstrings. *Arch Phys Med Rehabil*. 1996;77(7):688-692.
11. Holcomb WR. Improved stretching with proprioceptive neuromuscular facilitation. *J Natl Strength Cond Assoc*. 2000; 22(1):59-61.
12. Moore MA, Kukulka CG. Depression of Hoffmann reflexes following voluntary contraction and implications for proprioceptive neuromuscular facilitation therapy. *Phys Ther*. 1991;71(4):321-329.
13. Moore MA. Electromyographic investigation of muscle stretching techniques. *Med Sci Sports Exerc*. 1980;12:322-329.
14. Sady SP, Wortman M, Blanke D. Flexibility training: ballistic, static, or proprioceptive neuromuscular facilitation? *Arch Phys Med Rehabil*. 1982;63(6):261-263.
15. Sherrington C. *The Integrative Action of the Nervous System*. New Haven, CT: Yale University Press; 1947.
16. Wang RY. Effect of proprioceptive neuromuscular facilitation on the gait of patients with hemiplegia of long and short duration. *Phys Ther*. 1994;74(12):1108-1115.
17. Bachrach RM. Psoas dysfunction/insufficiency, sacroiliac dysfunction and low back pain. In: Vleeming A, Mooney V, Dorman T, Snijders C, Stoeckart R, eds. *Movement, Stability and Low Back Pain*. London: Churchill Livingstone; 1997: 309-317.
18. Cohen H. *Neuroscience for Rehabilitation*. 2nd ed. Philadelphia: Lippincott Williams & Wilkins; 1999.
19. Liebenson C. Active rehabilitation protocols. In: Liebenson C, ed. *Rehabilitation of the Spine*. Baltimore: Williams & Wilkins; 1996:355-390.
20. Milner-Brown A. *Neuromuscular Physiology*. Thousand Oaks, CA: National Academy of Sports Medicine; 2001.
21. Fox SI. *Human Physiology*. 5th ed. Dubuque, IA: Wm C Brown Publishers; 1996.
22. Vander A, Sherman J, Luciano D. *Human Physiology: The Mechanisms of Body Function*. 8th ed. New York: McGraw-Hill; 2001.
23. Enoka RM. *Neuromechanical Basis of Kinesiology*. 2nd ed. Champaign, IL: Human Kinetics; 1994.
24. McClosky DJ. Kinesthetic sensibility. *Physiol Rev*. 1978;58:763-820.
25. Grigg P. Peripheral neural mechanisms in proprioception. *J Sports Rehab*. 1994;3:2-17.
26. Janda V. Muscles and cervicogenic pain syndromes In: Grant R, ed. *Physical Therapy of the Cervical and Thoracic Spine*. Edinburgh: Churchill Livingstone; 1988:153-166.
27. Lewitt K. *Manipulation in Rehabilitation of the Locomotor System*. London: Butterworth; 1993.
28. Leahy PM. Active release techniques: Logical soft tissue treatment. In: Hammer WI, ed. *Functional Soft Tissue Examination and Treatment by Manual Methods*. Gaithersburg, MD: Aspen Publishers; 1999:549-559.
29. Spencer AM. *Practical Podiatric Orthopedic Procedures*. Cleveland: Ohio College of Podiatric Medicine; 1978.
30. Woo SLY, Buckwalter JA. *Injury and Repair of the Musculoskeletal Soft Tissues*. Park Ridge, IL: American Academy of Orthopedic Surgeons; 1987.
31. Zairns B. Soft tissue injury and repair—biomechanical aspects. *Int J Sports Med*. 1982;3:9-11.
32. Beaulieu JA. Developing a stretching program. *Physician Sports Med*. 1981;9:59-69.
33. Evjenth O, Hamburg J. *Muscle Stretching in Manual Therapy—A Clinical Manual*. Alfta, Sweden: Alfta Rehab; 1984.

34. Tannigawa M. Comparison of the hold-relax procedure and passive mobilization on increasing muscle length. *Phys Ther.* 1972;52:725-735.
35. Voss DE, Ionla MK, Meyers BJ. *Proprioceptive Neuromuscular Facilitation.* 3rd ed. Philadelphia: Harper and Row; 1985.
36. Akeson WH, Woo SLY. The connective tissue response to immobility: biochemical changes in periarticular connective tissue of the immobilized rabbit knee. *Clin Orthop Relat Res.* 1973;93:356-362.
37. Barnes JF. Myofascial release. In: Hammer WI, ed. *Functional Soft Tissue Examination and Treatment by Manual Methods.* 2nd ed. Gaithersburg, MD: Aspen Publishers; 1999:533-547.
38. Sapega A, Quedenfeld T, Moyer R. Biophysical factors in range of motion exercises. *Phys Sports Med.* 1981;9:57-65.
39. Etnyre BR, Abraham LD. Gains in range of ankle dorsiflexion using three popular stretching techniques. *Am J Phys Med.* 1986;65:189-196.